SOCI 502 Data Story

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MILESTONE 1: QUESTION AND SIMULATION

Research Question:

Do Canadian men and women choose different commuting modes and if so, does this gender difference vary by province/territory?

Dataset:

The dataset that I have selected is Table 98-10-0458-01 (Government of Canada 2021) from the 2021 Canadian Census. This dataset includes information on Canadians' choice of commuting mode, duration of commute, time leaving for work, age, and gender, categorized by province/territory.

Plans for Data Story:

I plan to compare transit use of Canadian men and women across all 13 provinces and territories in Canada to see if there are patterns that can be identified, when comparing transit use with total commuters in each province.

I hypothesize that women across each province will have higher transit use rates than their male counterparts, and that the provinces with the highest rates of transit use will be large provinces with sophisticated transit networks. This will be proven if Women+ transit rates across the 13 Canadian provinces and territories are greater than those of Men+ and if transit rates are highest within "large_tier" provinces and lowest within "small_tier" provinces.

Simulated Code and Data:

```
#### Workspace Setup ####

# install.packages("tidyverse")
# install.packages("dplyr")
# install.packages("knitr")
# install.packages("tinytable")

library(tidyverse)
library(dplyr)
library(knitr)
library(tinytable)
```

```
set.seed(723) # makes the simulation reproducible

# Naming simulated objects (note that "provinces" includes both Canadian
# provinces and territories for simplicity)

provinces <- c(
  "Newfoundland and Labrador", "Prince Edward Island", "Nova Scotia",
  "New Brunswick", "Quebec", "Ontario", "Manitoba", "Saskatchewan", "Alberta",
  "British Columbia", "Yukon", "Northwest Territories", "Nunavut"
)
```

```
# Rough relative size of each workforce (approx. in thousands, per gender)

workforce_k <- c(90, 35, 220, 175, 2000, 3500, 290, 240, 1000, 1100, 10, 10, 10)

# Simulating provinces' workforce numbers by gender with ±10% difference

sim_grid <- expand_grid(geography = provinces, gender = c("Men+", "Women+")) |>
  mutate(
    total = round(rep(workforce_k, each = 2) * 1000 * runif(26, 0.9, 1.1))
  )

sim_grid
```

```
# A tibble: 26 x 3
   geography                 gender   total
   <chr>                     <chr>    <dbl>
```

```
 1 Newfoundland and Labrador Men+     89529
 2 Newfoundland and Labrador Women+   96500
 3 Prince Edward Island      Men+     35118
 4 Prince Edward Island      Women+   37654
 5 Nova Scotia               Men+    208050
 6 Nova Scotia               Women+  234297
 7 New Brunswick             Men+    167002
 8 New Brunswick             Women+  189127
 9 Quebec                    Men+   1815278
10 Quebec                    Women+ 1921510
# i 16 more rows
```

```r
# Baseline probability of using transit, set in three simple tiers (based on
# expectation, not estimates from data)

# Large transit systems (Quebec, Ontario, BC): 8%
# Medium transit systems (Nova Scotia, Manitoba, Alberta): 4%
# All other provinces and territories: 2%

large_tier <- c("Quebec", "Ontario", "British Columbia")
medium_tier <- c("Nova Scotia", "Manitoba", "Alberta")
small_tier <- c("Yukon", "Northwest Territories", "Nunavut",
                "Prince Edward Island", "Newfoundland and Labrador",
                "Saskatchewan", "New Brunswick")

base_p <- case_when(
  provinces %in% large_tier  ~ 0.08,
  provinces %in% medium_tier ~ 0.04,
  provinces %in% small_tier  ~ 0.02
)

base_logit <- qlogis(base_p) # convert to the log-odds scale used by the model
```

```r
# Gender effect on the log-odds scale: women are assumed to be higher on average
# (+0.25), and the size of the gap varies randomly by province.
# This variation is the interaction I want to detect.

gender_effect <- 0.25 + rnorm(13, mean = 0, sd = 0.15)

sim_grid <- sim_grid |>
  mutate(
    province_id = match(geography, provinces),
```

Province	Gender	Transit Users	Total Commuters	Share
Newfoundland and Labrador	Men+	1,863	89,529	2.1Newfoundland and Labrador

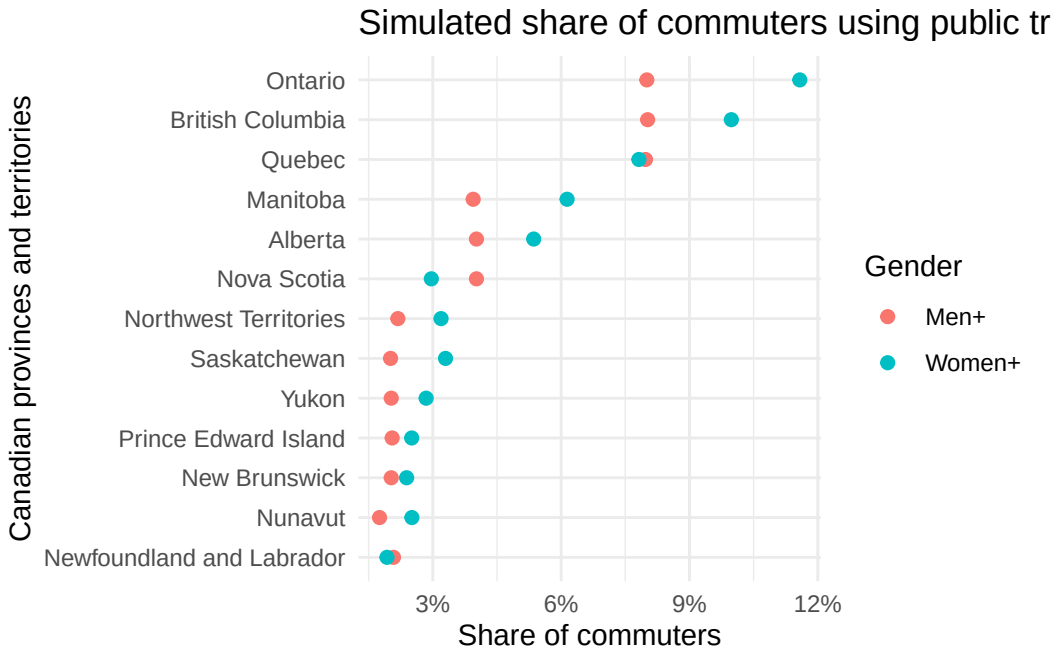
```
    logit_p = base_logit[province_id] +
      if_else(gender == "Women+", gender_effect[province_id], 0),
    p_transit = plogis(logit_p),
    transit = rbinom(n(), size = total, prob = p_transit),
    share = transit / total
    ) |>
  select(geography, gender, transit, total, share)

print(sim_grid)
```

```
# A tibble: 26 x 5
   geography                 gender transit   total  share
   <chr>                     <chr>    <int>   <dbl>  <dbl>
 1 Newfoundland and Labrador Men+      1863   89529 0.0208
 2 Newfoundland and Labrador Women+    1862   96500 0.0193
 3 Prince Edward Island      Men+       720   35118 0.0205
 4 Prince Edward Island      Women+     944   37654 0.0251
 5 Nova Scotia               Men+      8366  208050 0.0402
 6 Nova Scotia               Women+    6949  234297 0.0297
 7 New Brunswick             Men+      3388  167002 0.0203
 8 New Brunswick             Women+    4517  189127 0.0239
 9 Quebec                    Men+    144714 1815278 0.0797
10 Quebec                    Women+  150227 1921510 0.0782
# i 16 more rows
```

```
sim_grid |>
  tt() |>
  style_tt(j= 1:5, align = "llrrr")|>
  format_tt(j = 3:4, digits = 4, num_mark_big = ",", num_fmt = "decimal") |>
  format_tt(j = 5, fn = scales::label_percent(accuracy = 0.1)) |>
  setNames(c("Province", "Gender", "Transit Users", "Total Commuters", "Share"))
```

```
ggplot(sim_grid, aes(x = share, y = fct_reorder(geography, share),
                     colour = gender)) +
  geom_point (size = 2) +
  scale_x_continuous(labels = scales::percent) +
```

```
labs(title = "Simulated share of commuters using public transit",
     x = "Share of commuters", y = "Canadian provinces and territories",
     colour = "Gender") +
theme_minimal()
```



Government of Canada. 2021. "Main Mode of Commuting by Commuting Duration, Time Leaving for Work, Age and Gender: Canada, Provinces and Territories, Census Divisions and Census Subdivisions." https://doi.org/https://doi.org/10.25318/9810045801-eng.